DYNAMIC STOPPING DISTANCE: REACTION LAG + KINETIC BRAKING

$d\_stop(t) = v(t) \cdot \Delta t\_latency + v(t)^2 / (2\ a\_max) \leq d\_clearance$

1. SENSE-TO-ACTUATION REACTION DISTANCE

$$d\_reaction = v_0 \cdot \Delta t\_wall$$

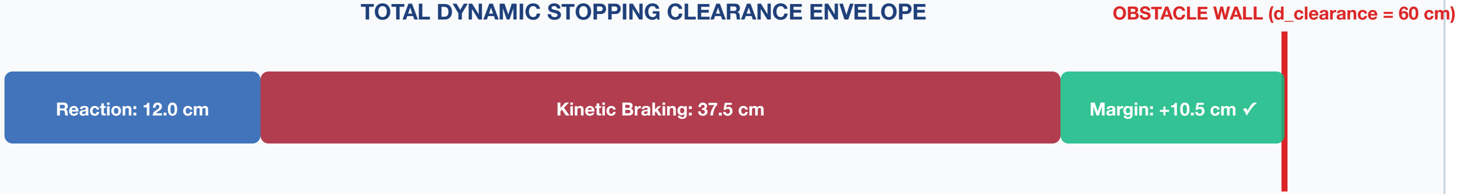
- Vehicle travels at initial speed during computation lag
- Linear growth with latency: doubling latency doubles distance
- Example: at 1.5 m/s, Δt=80 ms ⇒ d_reaction = 12.0 cm

2. PHYSICAL KINETIC BRAKING DISTANCE

$$d\_braking = v_0^2 / (2 \cdot a\_max)$$

- Dissipating kinetic energy $E\_k = 1/2\ m\ v^2$ against friction μ
- Quadratic growth with velocity: doubling speed quadruples distance
- Example: at 1.5 m/s, a_max=3.0 m/s² ⇒ d_braking = 37.5 cm

TOTAL DYNAMIC STOPPING CLEARANCE ENVELOPE



Total Stopping Distance: d_stop = 49.5 cm ≤ Clearance: 60.0 cm (Certified Safe)

If latency spikes to 200 ms (P99.9 tail), d_reaction becomes 30 cm ⇒ d_stop = 67.5 cm > 60 cm ⇒ COLLISION!